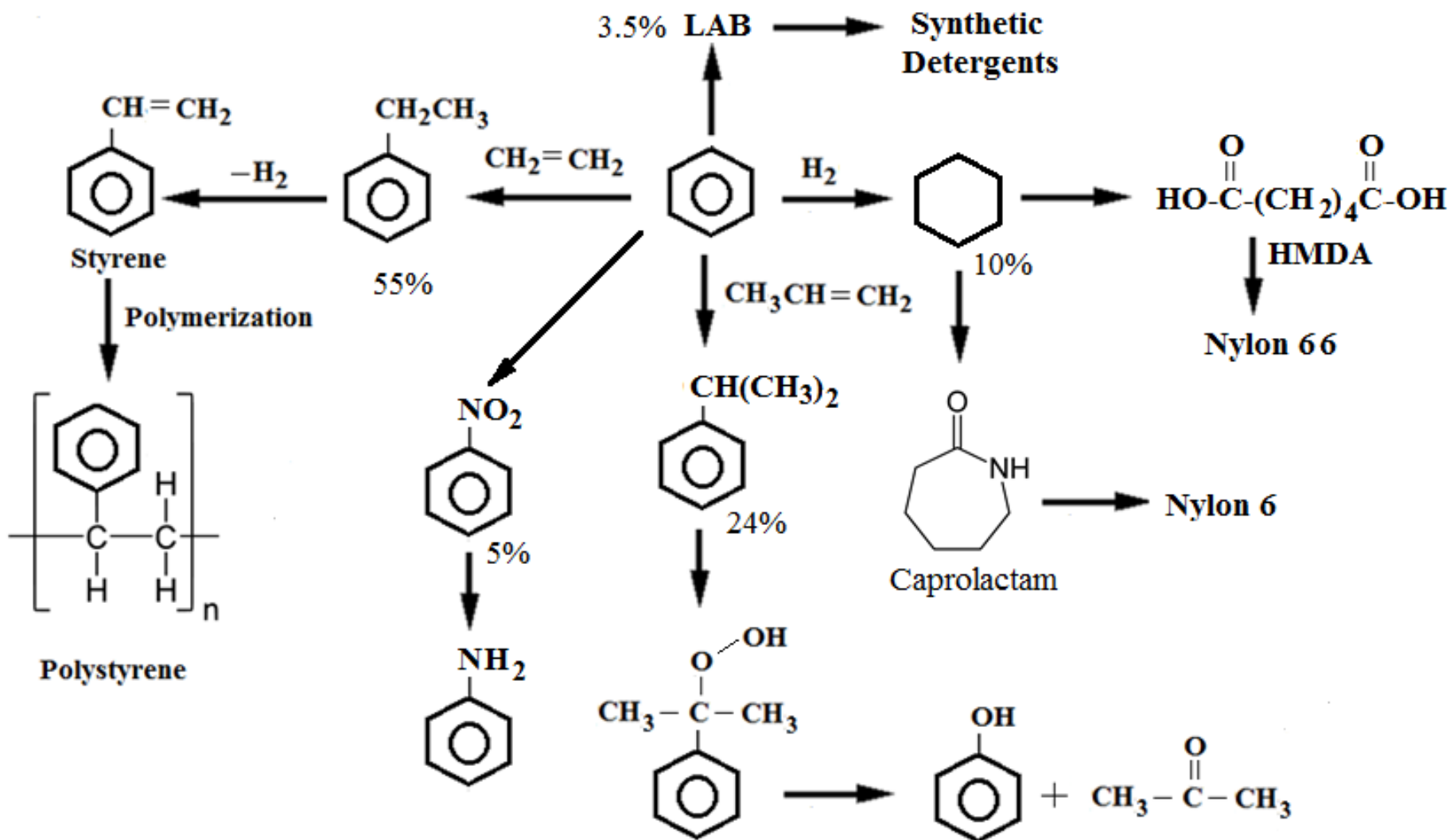


Petrochemical Technology

Chemicals from Benzene

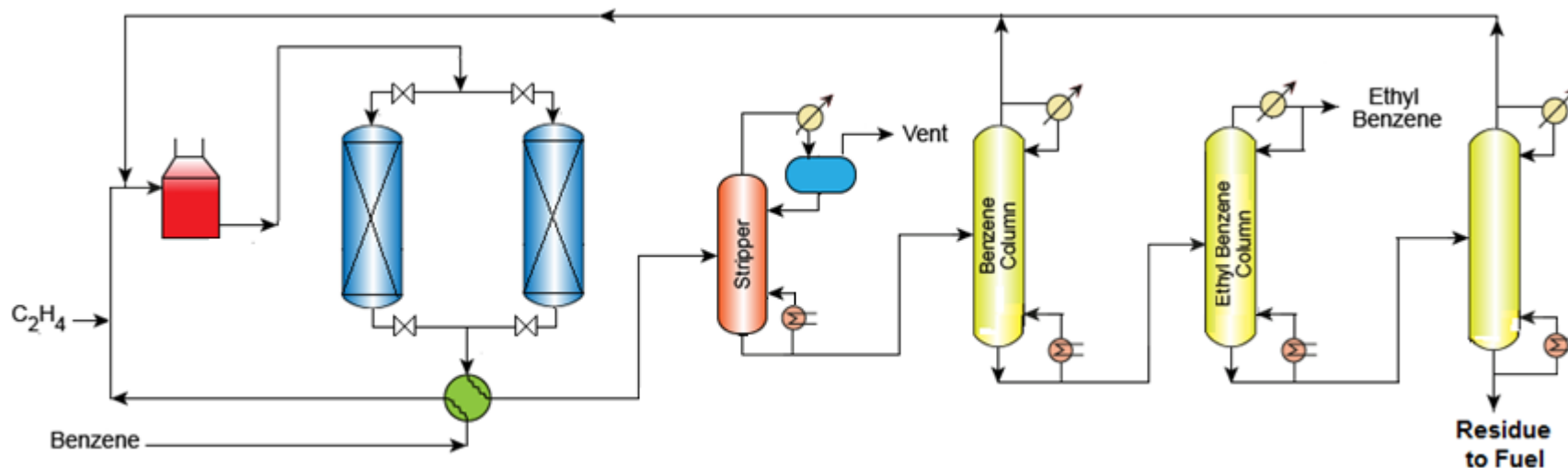
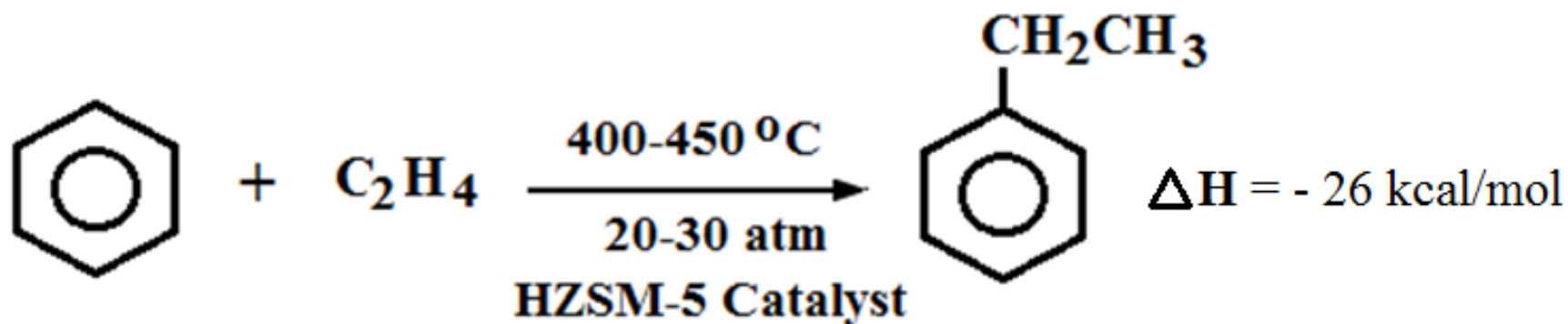
Major Chemicals from Benzene

Global Benzene demand stood at 52.94 Million Tonnes in 2020 and is forecast to reach 78.03 Million Tonnes by 2030, growing at a healthy CAGR of 3.81% until 2030.

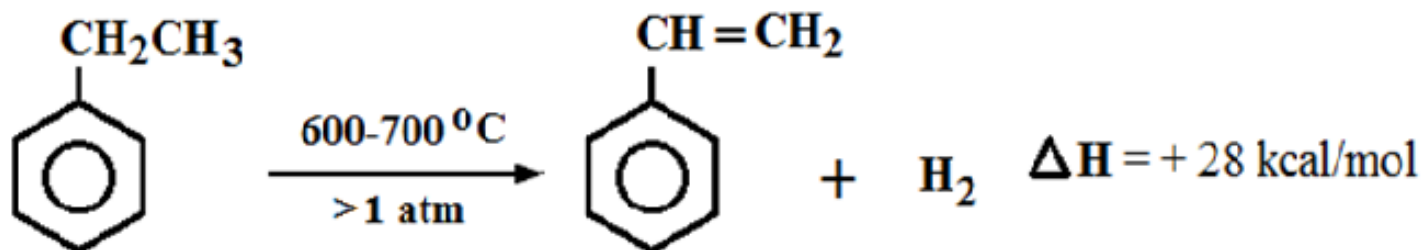


Production of Ethylbenzene and Styrene

Mobil-Badger Process for Ethylbenzene

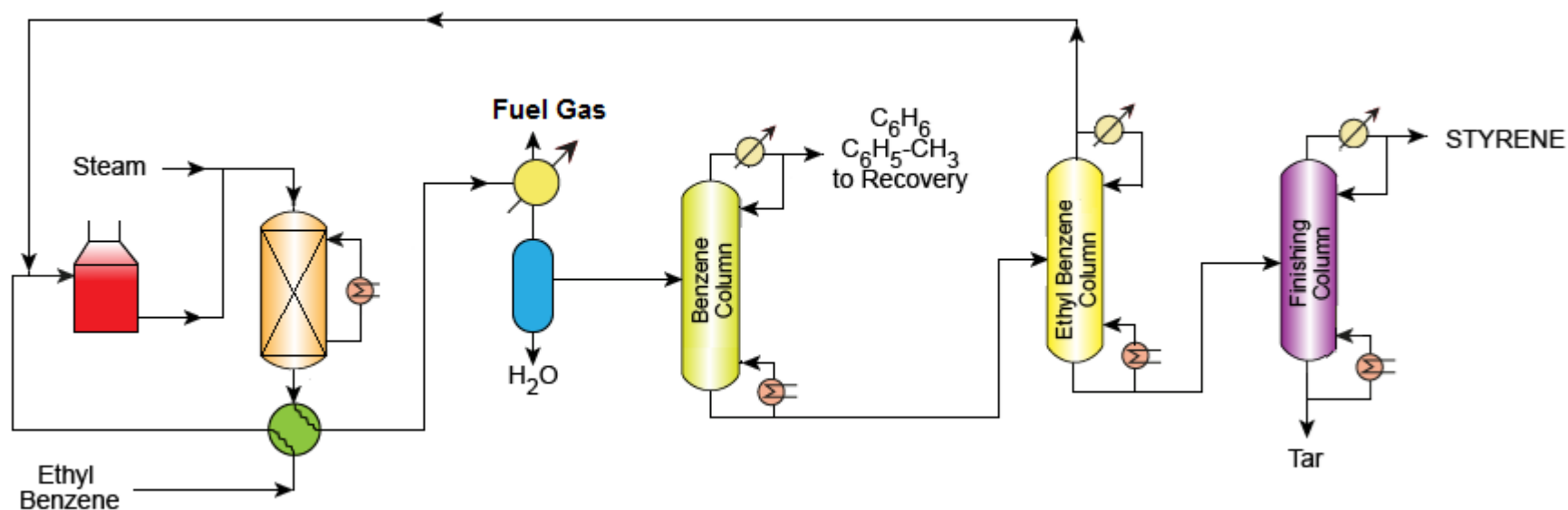


Production of Ethyl Benzene

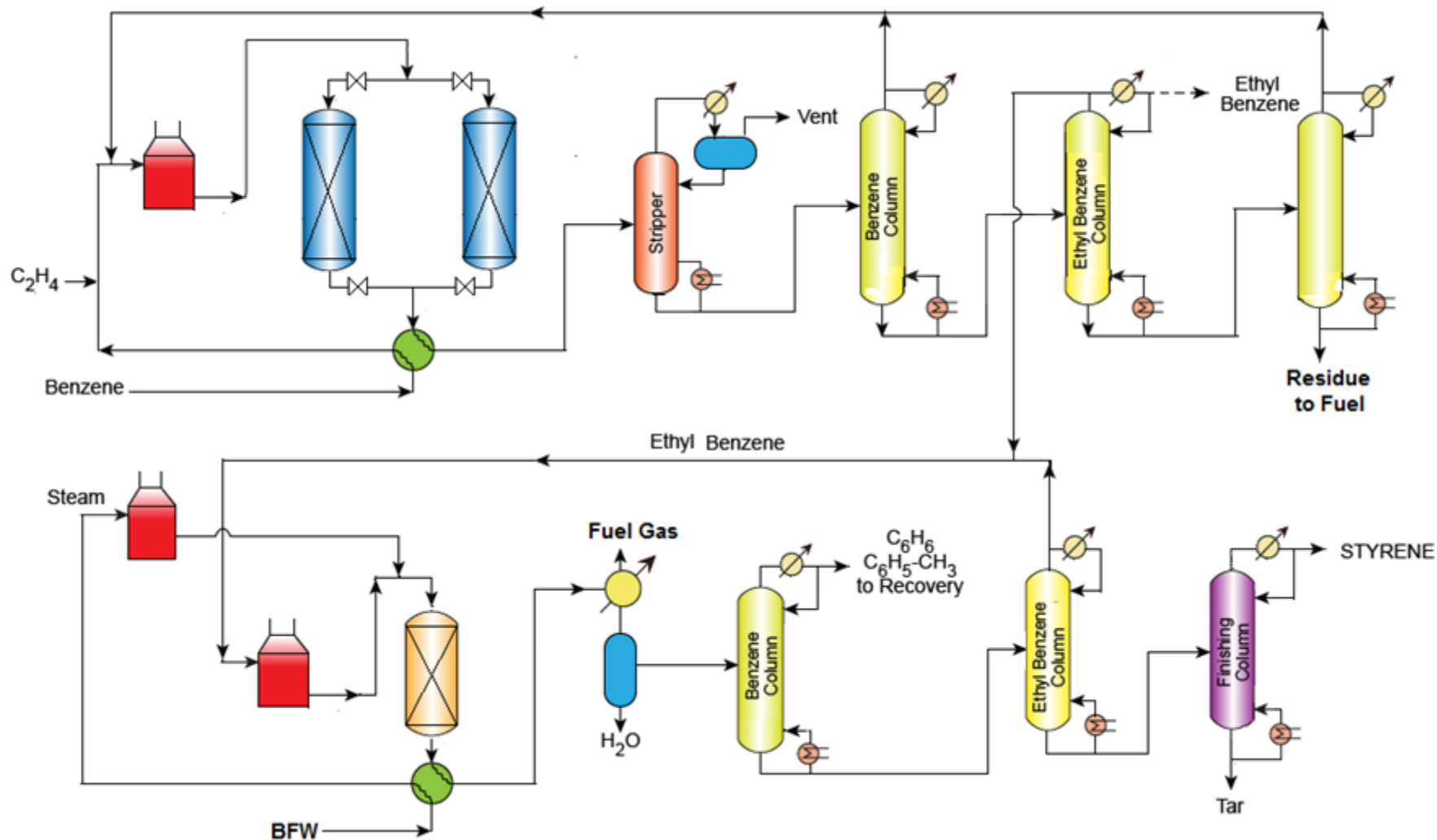


Catalyst: Fe_2O_3 or ZnO mixed with 0.5-1% Cr_2O_3

All these catalysts are mixed with a small amount of potassium carbonate which facilitates the removal of carbon deposited on the catalyst surface, thereby prolonging its activity.

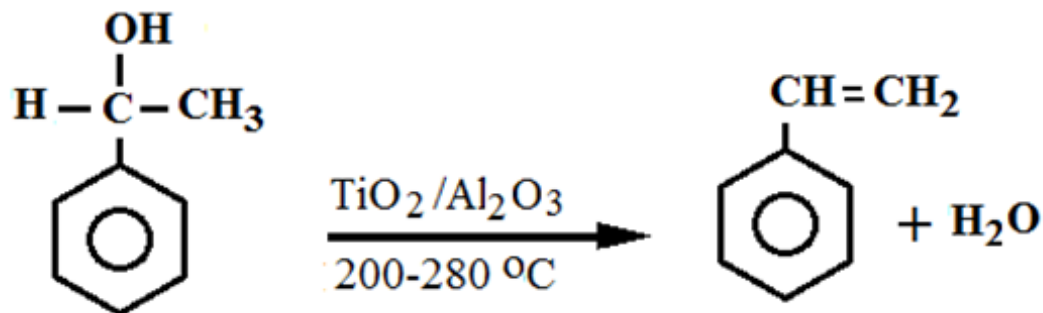
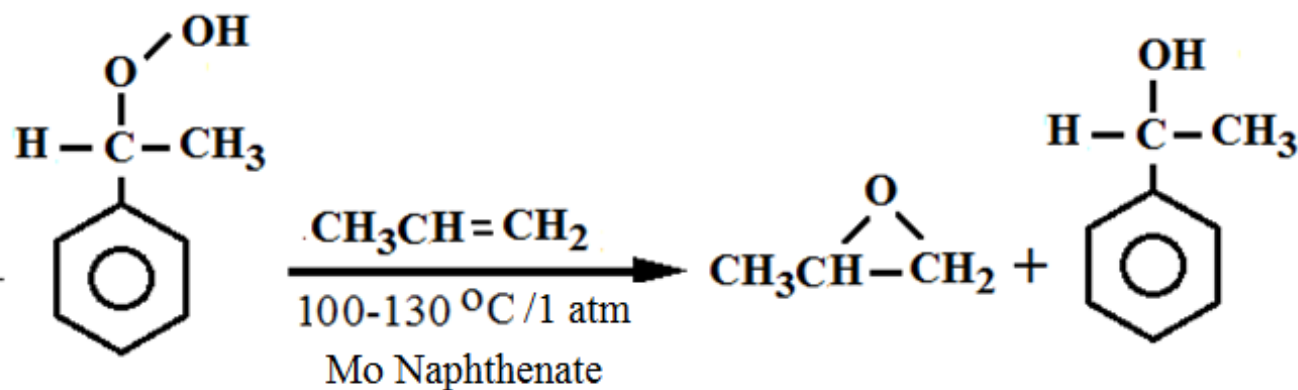
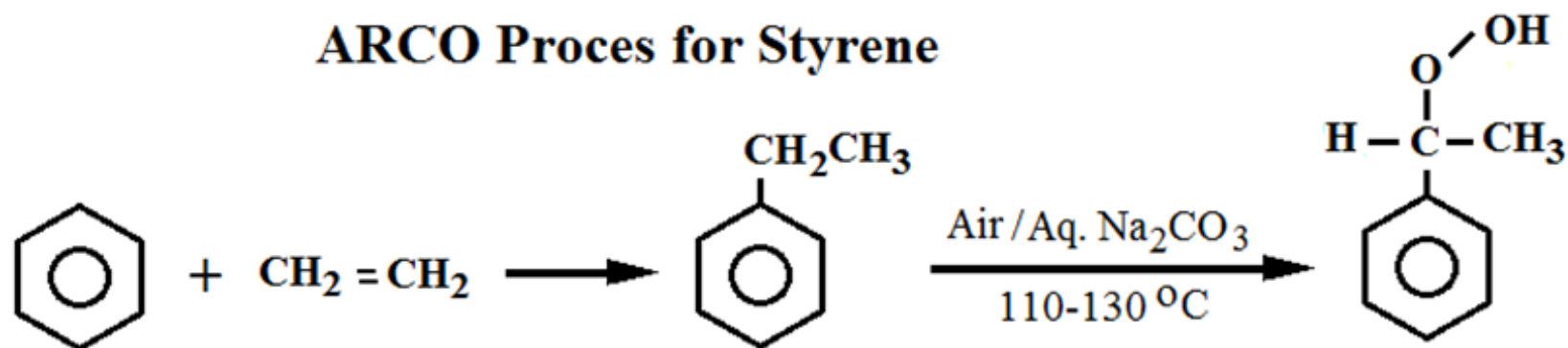


Production of Styrene from Ethyl Benzene



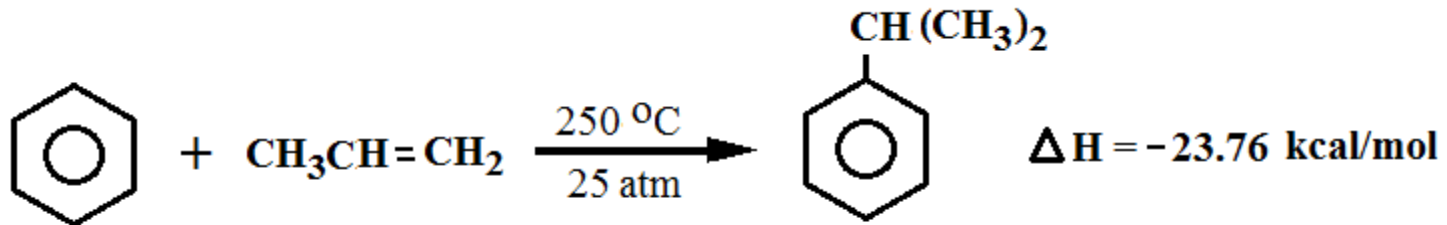
Production of Styrene from Benzene

ARCO Proces for Styrene

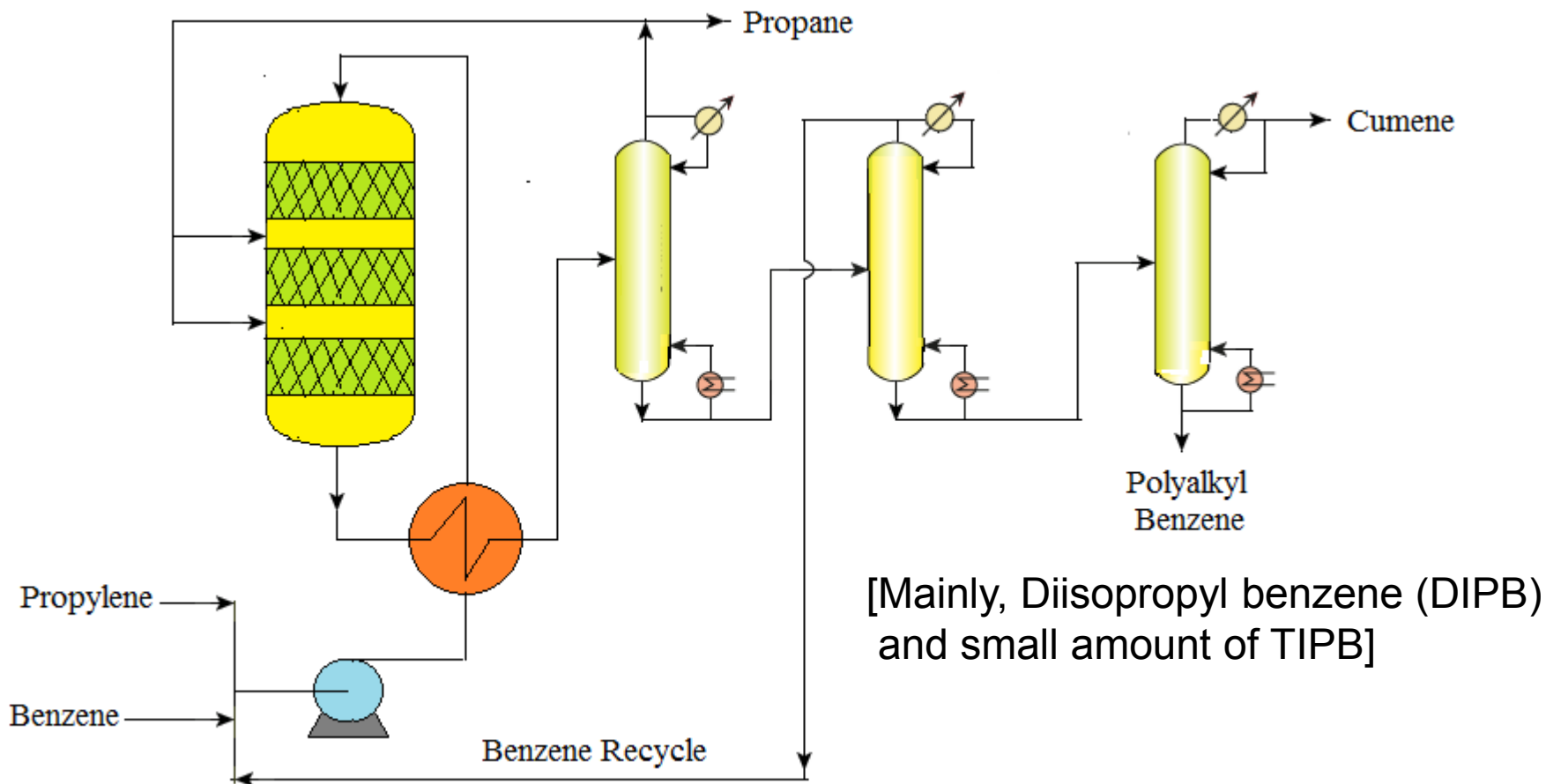


This is also a route to propylene oxide by indirect oxidation of propylene

Production of Cumene and Phenol

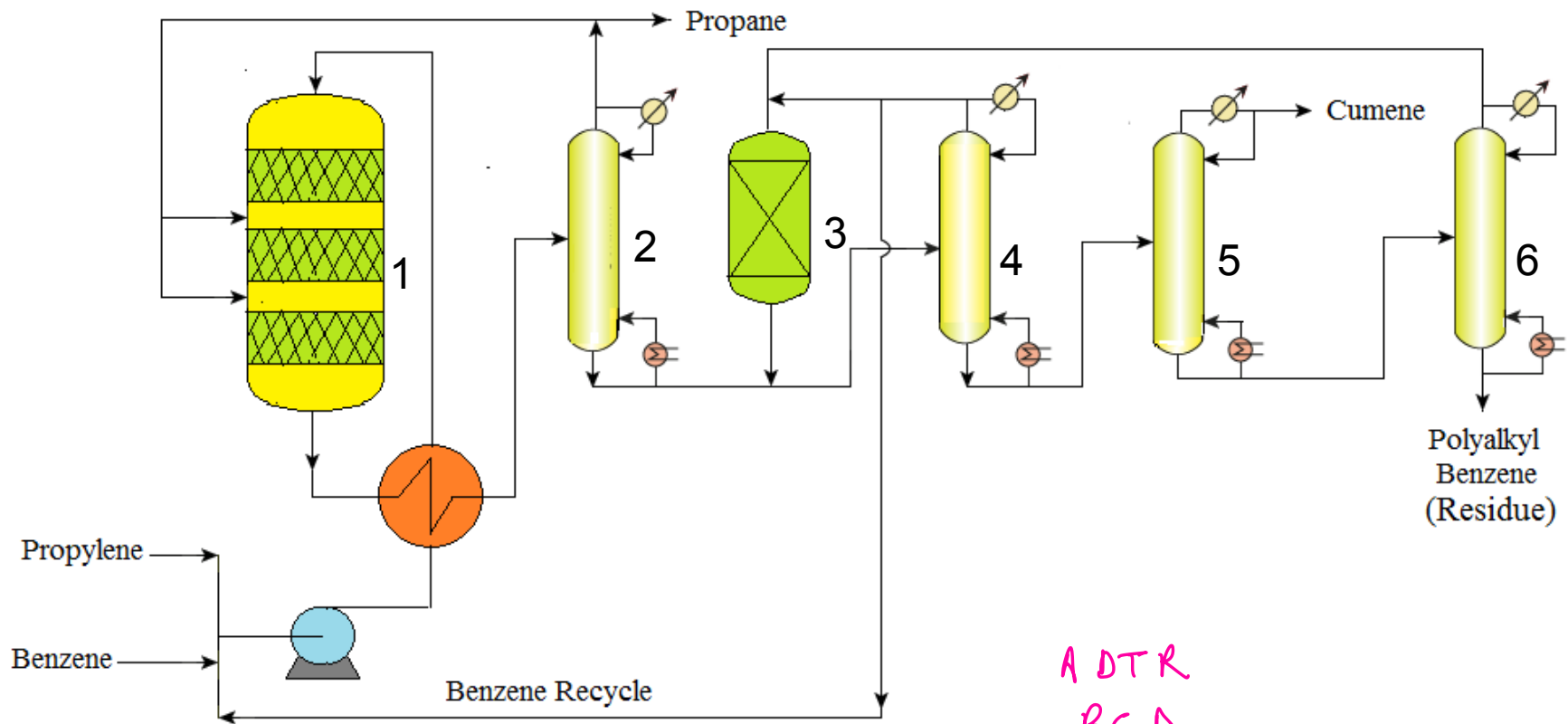


Catalyst: Solid phosphoric acid (Phosphoric acid on Kieselguhr)



Production of Cumene by Propylation of Benzene

Catalyst: QZ-2001



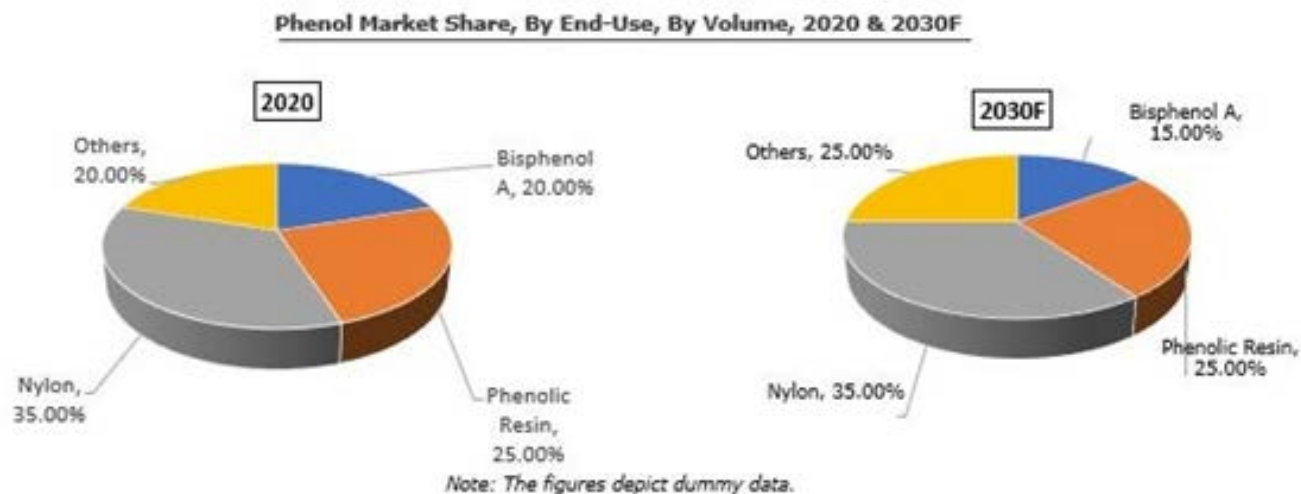
1: Alkylation Reactor, 2: Depropanizer, 3: Transalkylation Reactor,
4: Benzene Column, 5: Cumene Column, 6: DIPB Column

UOP's Q-Max Process for Cumene

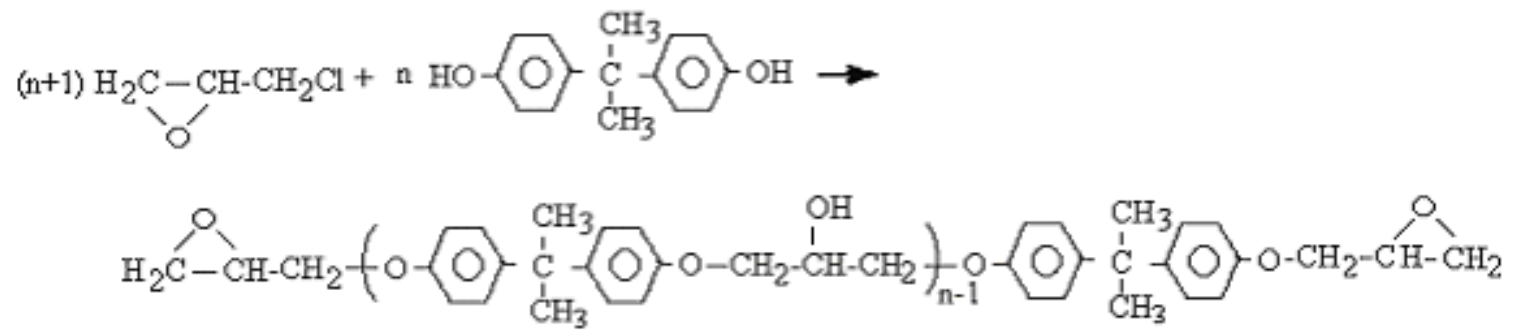
In fiscal year 2021, the production volume of phenol across India was around 61 thousand metric tons.

The demand for Phenol in India reached 311.00 thousand tonnes in FY2021 and is anticipated to grow at around CAGR 5.20% to reach 490.80 thousand tonnes by FY2030.

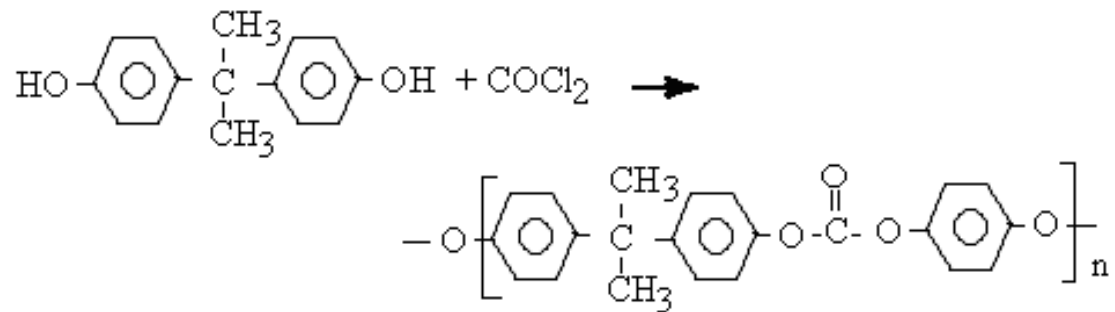
Globally Phenol is consumed majorly in BPA production. Major applications for BPA come from polycarbonate and epoxy resins. Polycarbonates are widely used in electronics, automotive, optical, etc. industries whereas epoxy resins are widely used in the paint & coating industry. Phenol is also used in producing Phenolic / Formaldehyde resin, which is further used in furniture, laminates, etc.

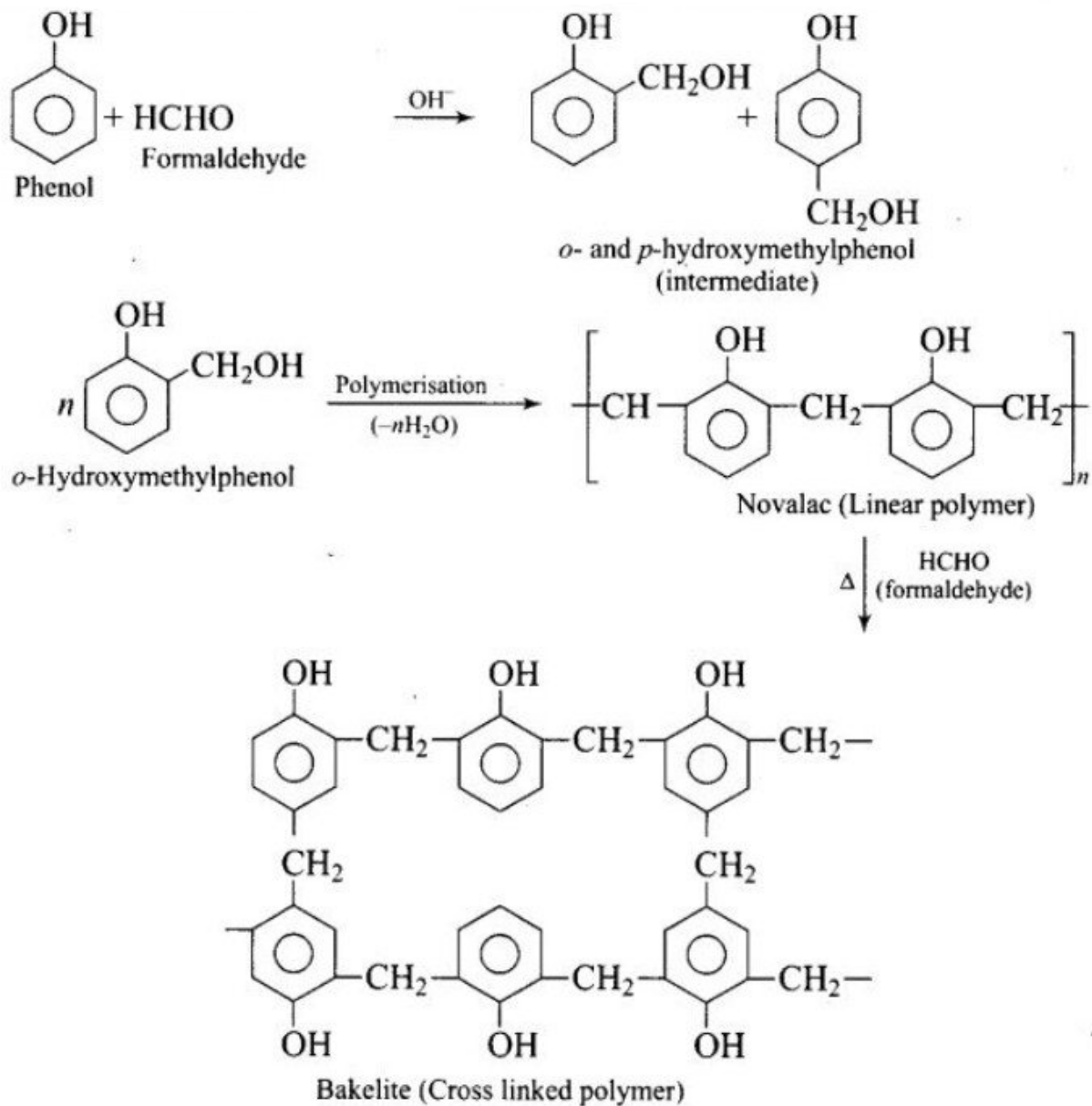


Epoxy Resin

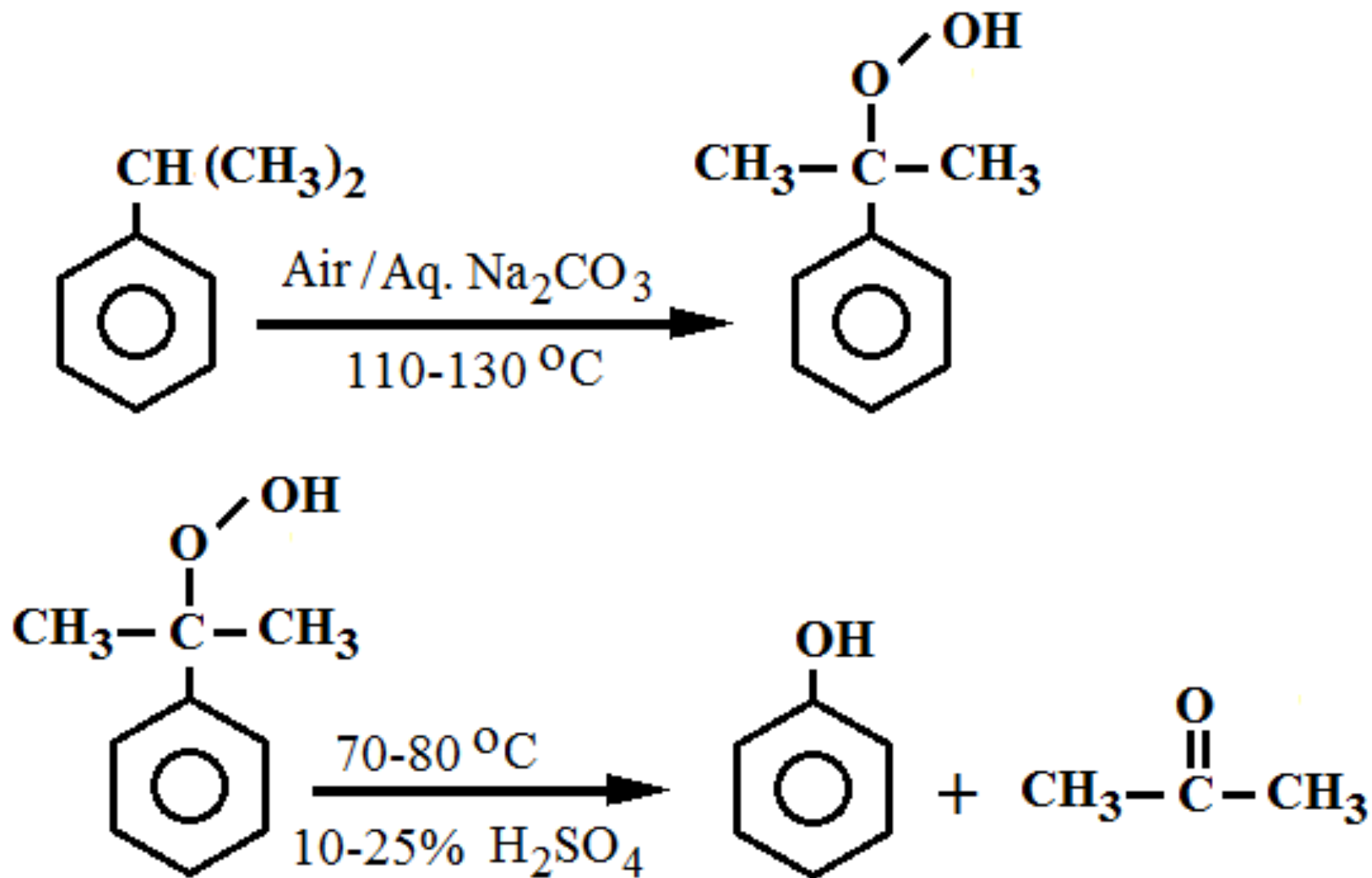


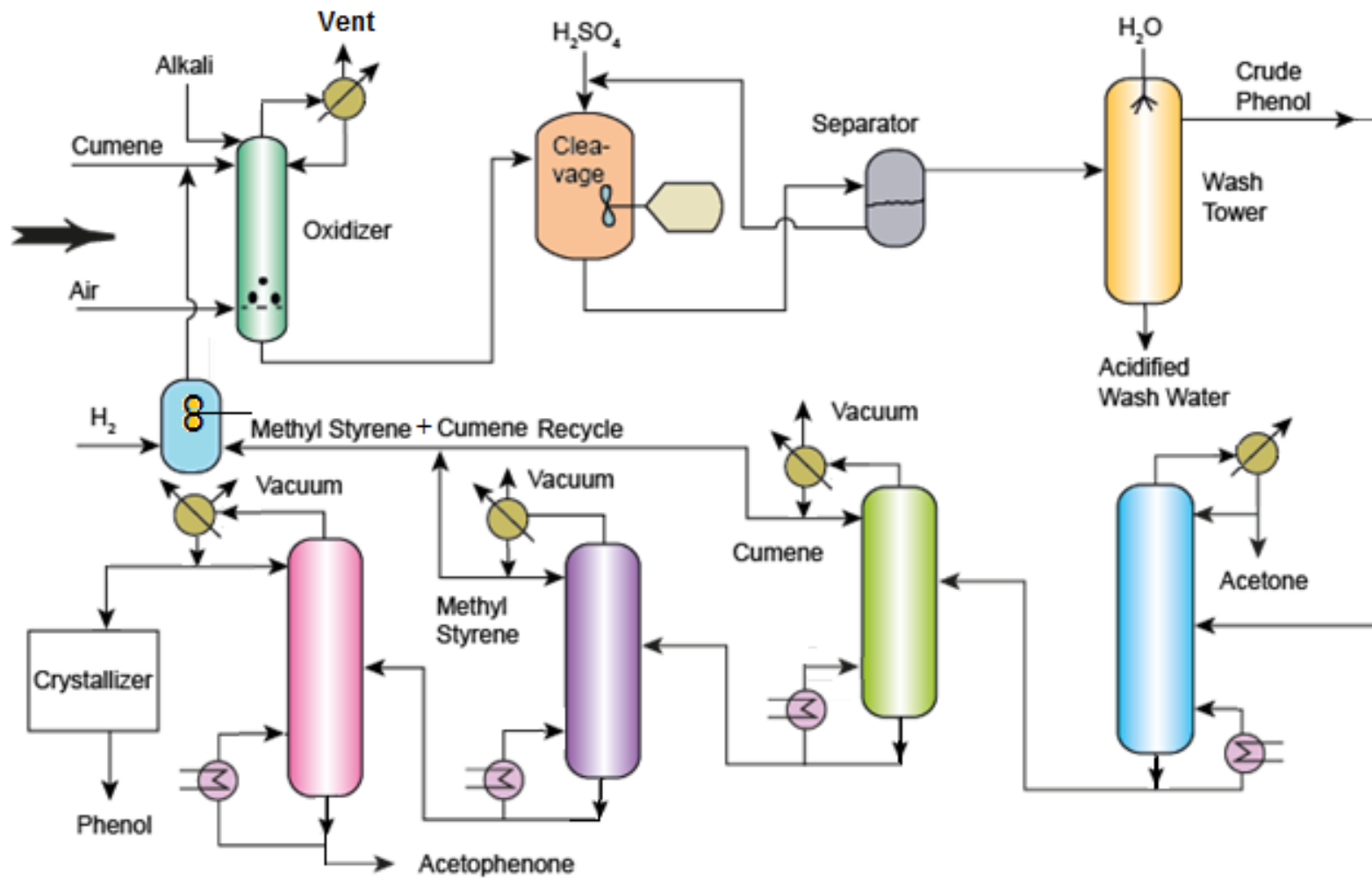
Polycarbonate Resin





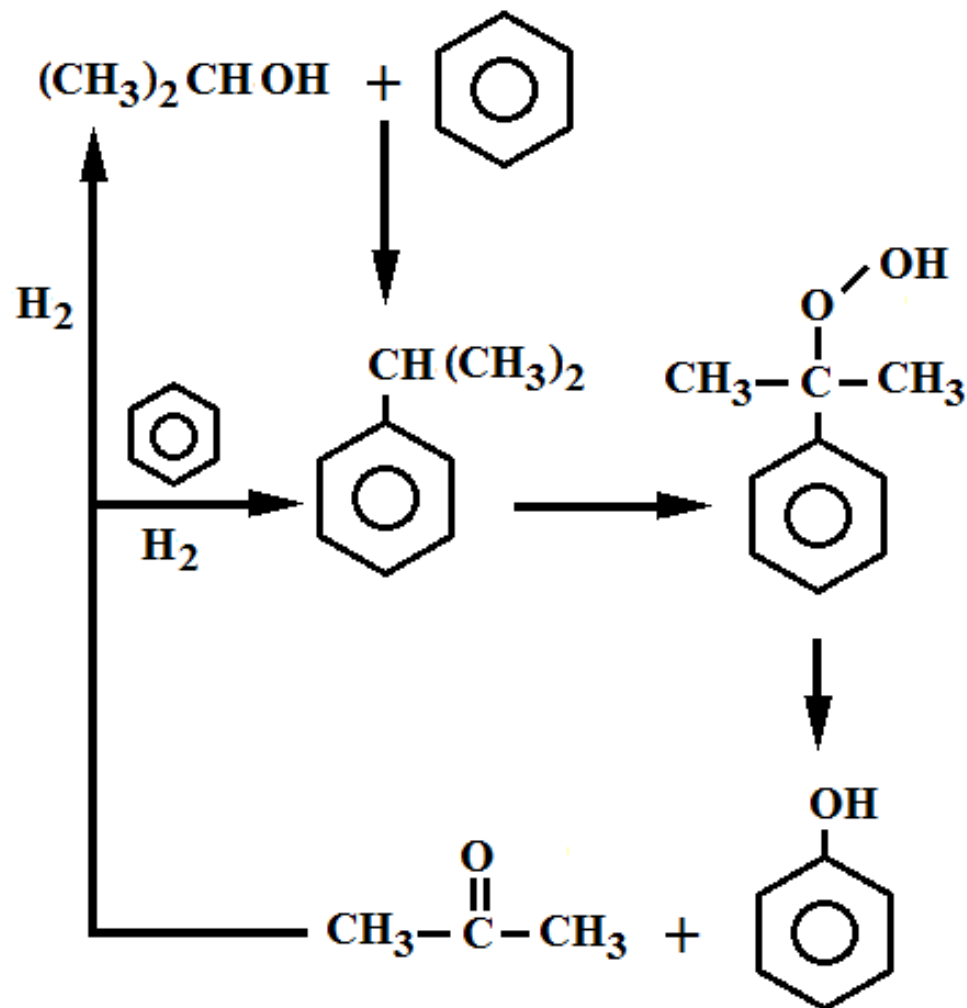
Synthesis of Phenol from Cumene





Production of Phenol from Cumene

Cumene synthesis using by-product Acetone



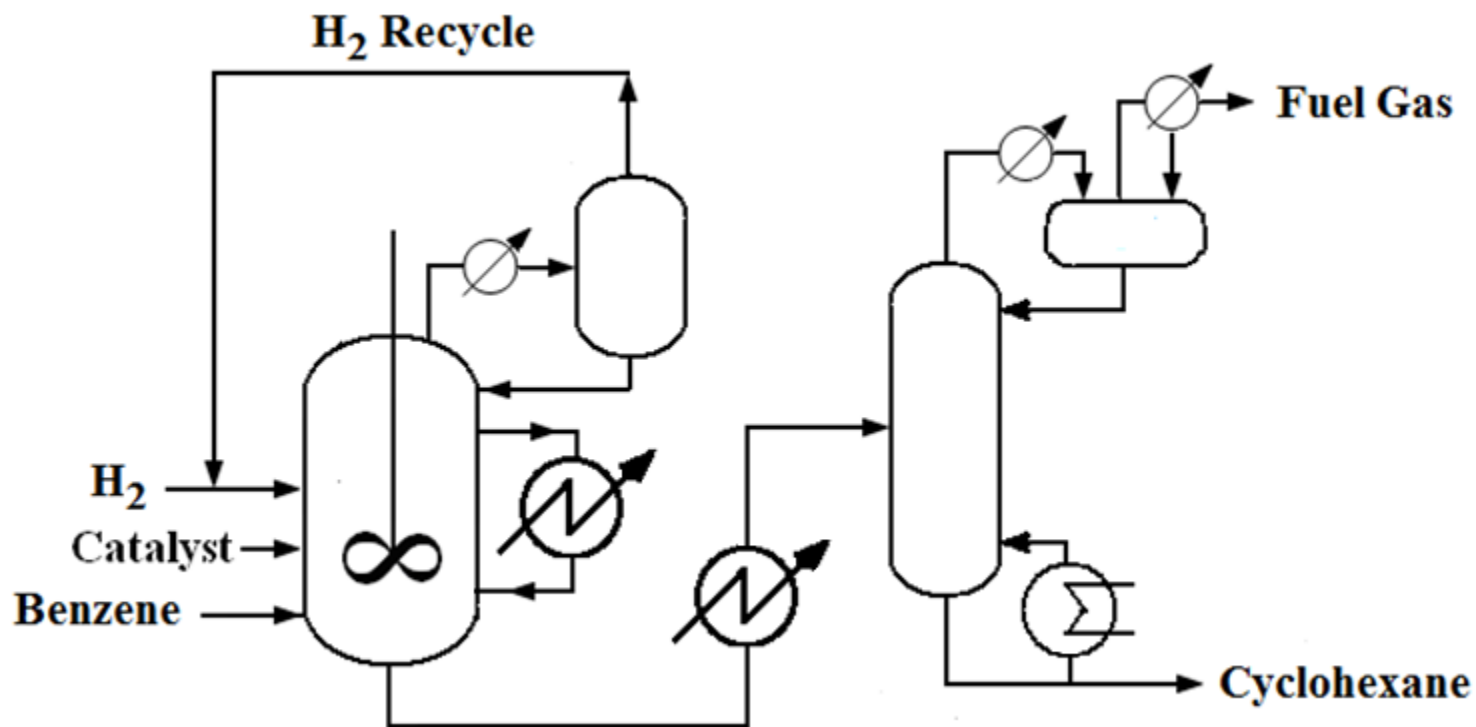
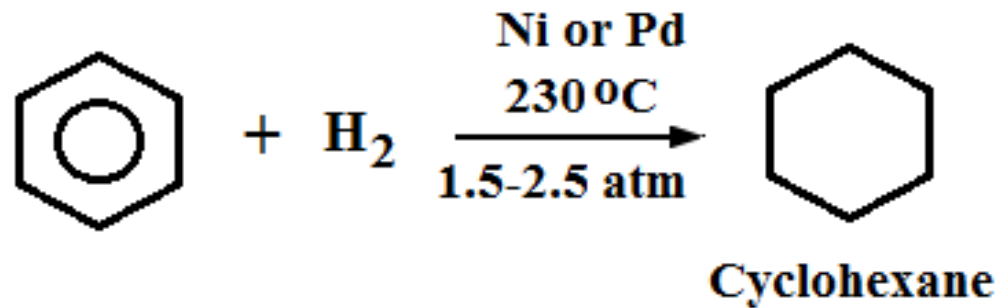
Sreedevi *et al.*; *Ind. Engg. Chem. Res.* **2001**, 40(14), 3133-3138

Barman *et al.*; *Ind. Engg. Chem. Res.* **2006**, 45(10), 3481-3487

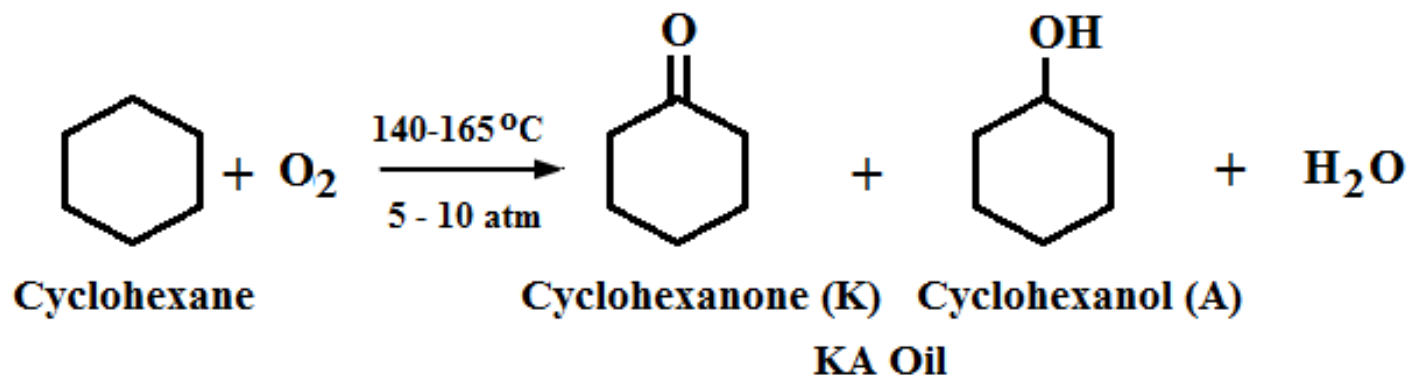
Basu, S. and Pradhan, N. C.; *Catal. Today*, **2019**, 348, 118-126.

Production of Adipic Acid and Caprolactam

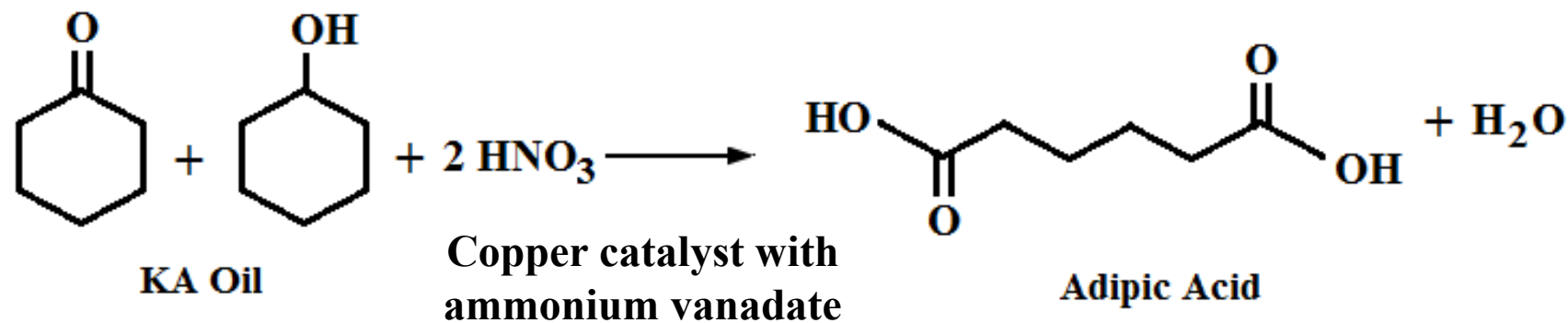
Cyclohexane from Benzene

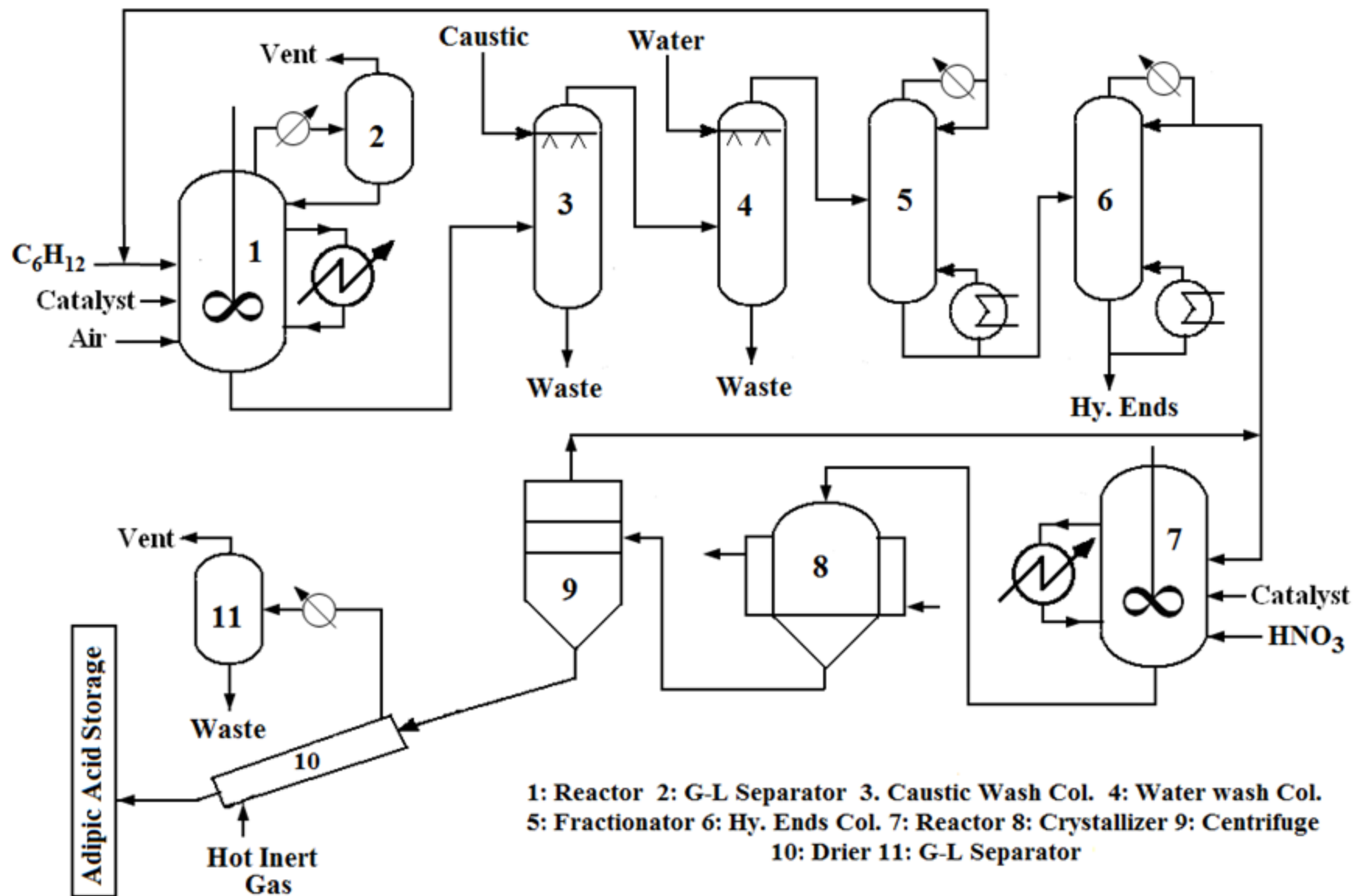


Adipic Acid from Cyclohexane

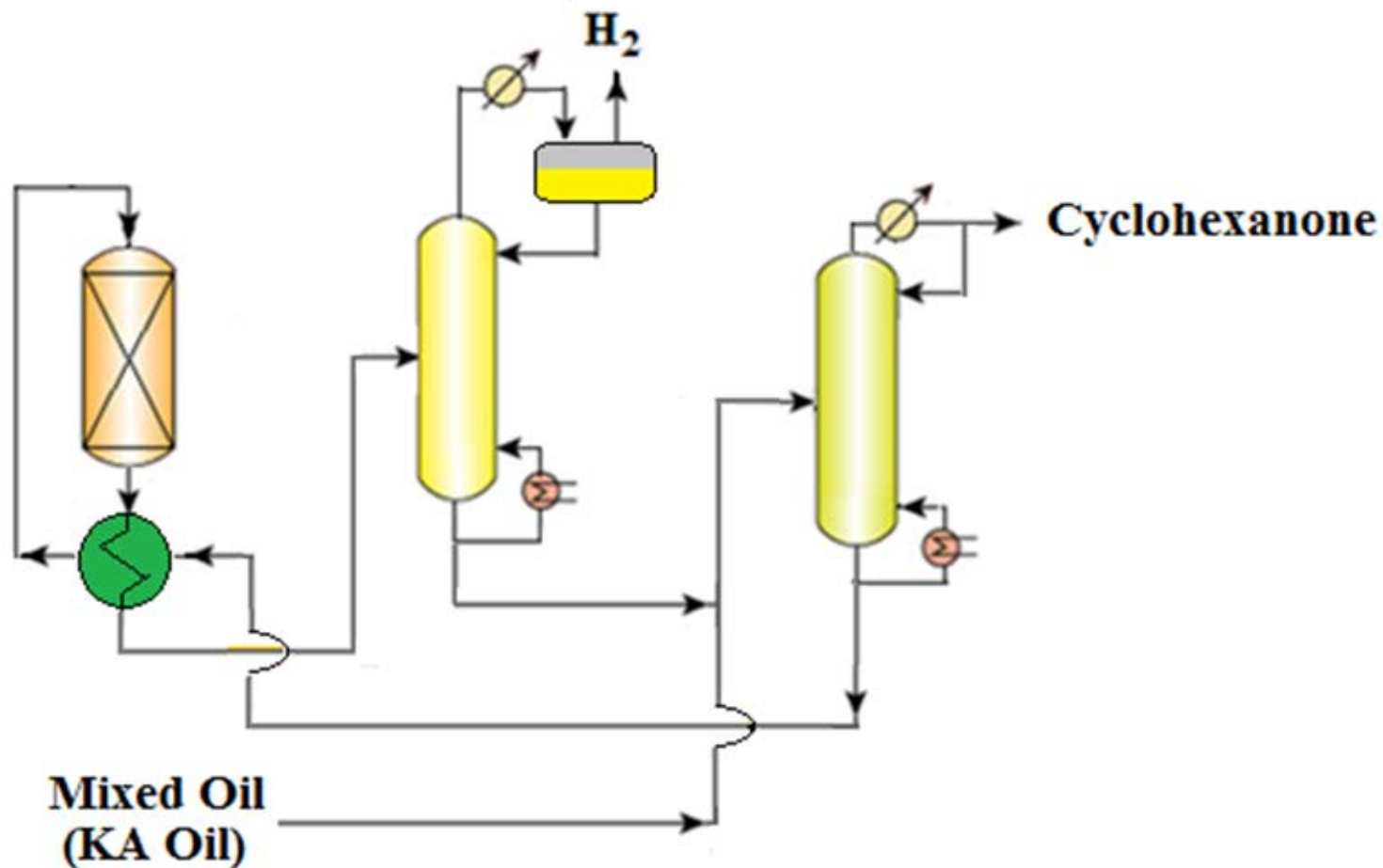
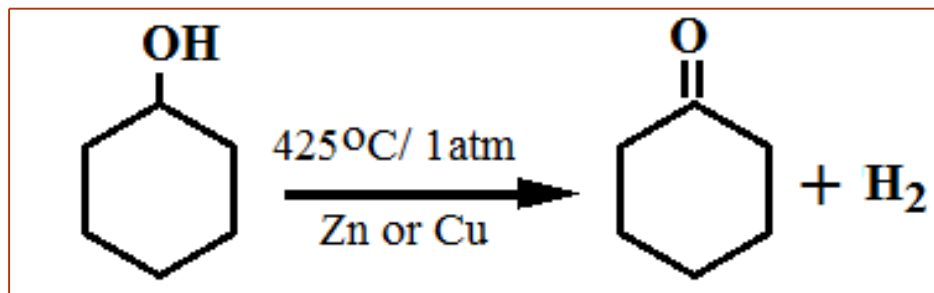


Catalyst: Cobalt salt (Cobalt naphthenate or octanoate)

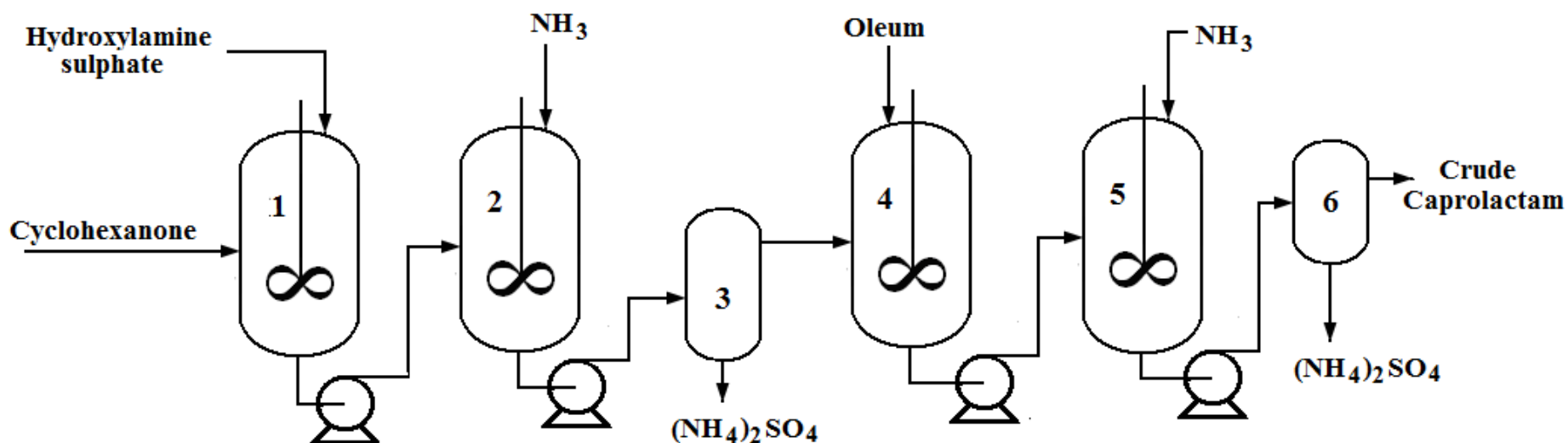
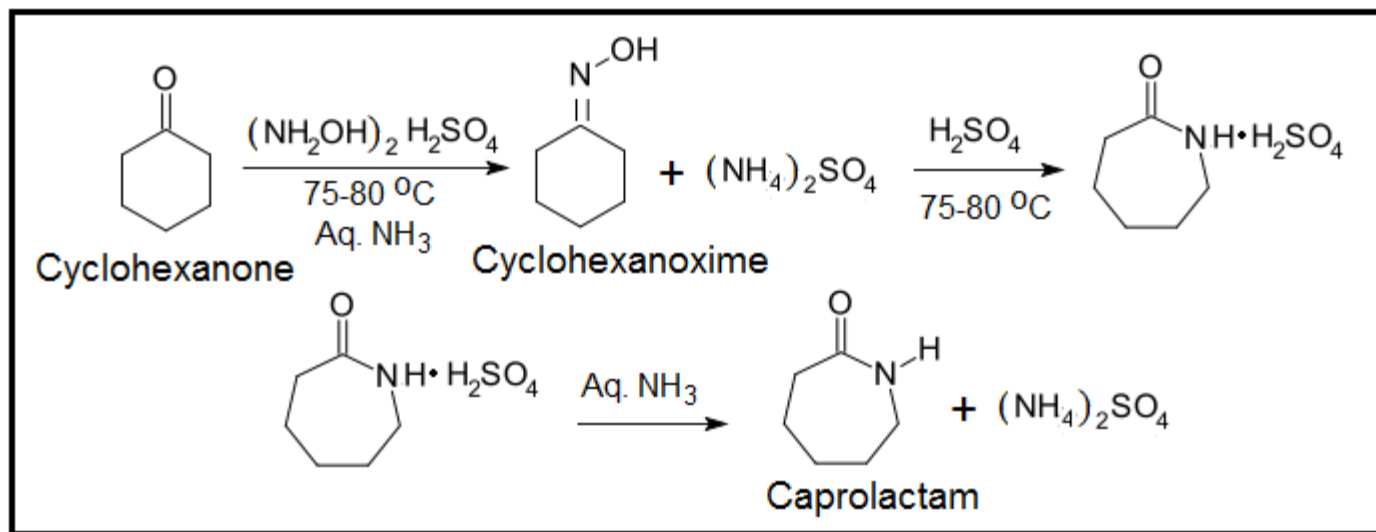




Production of Adipic Acid from Cyclohexane



Production of Cyclohexanone from Cyclohexanol



1: Oximation Reactor; 2 & 5 Neutralizer; 3 & 6: Settler
4: Reactor for Beckmann Rearrangement

Caprolactam from Cyclohexanone

Production of **Nitrobenzene** **and** **Aniline**

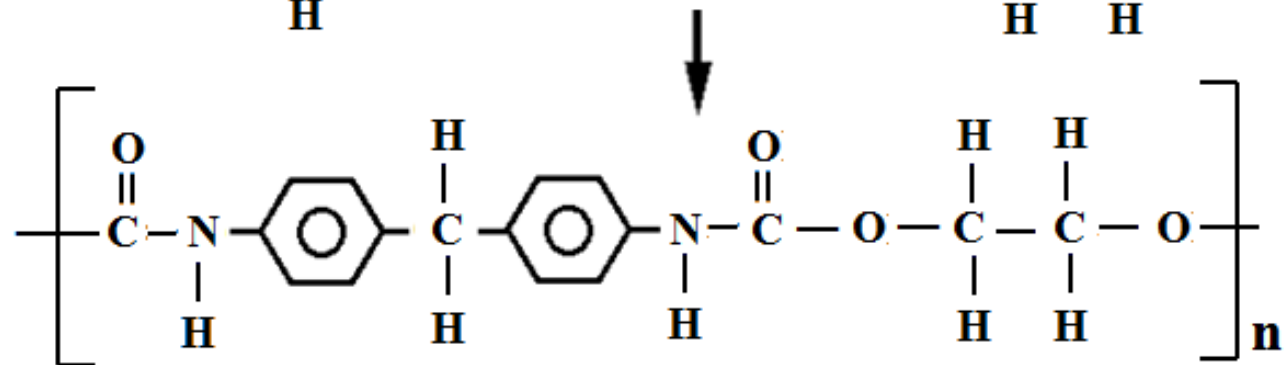
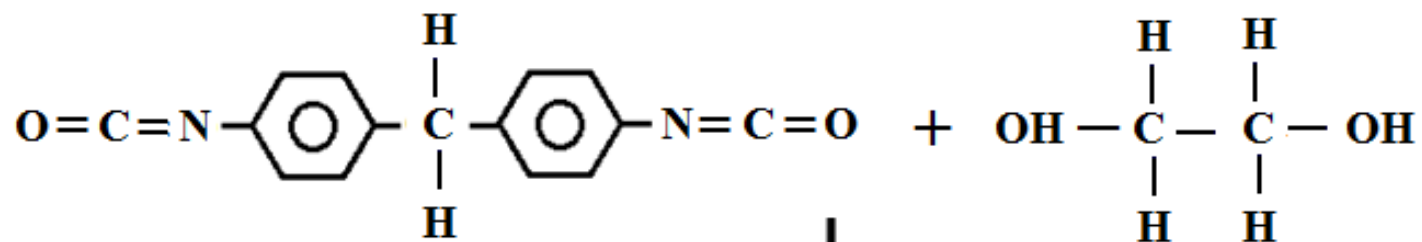
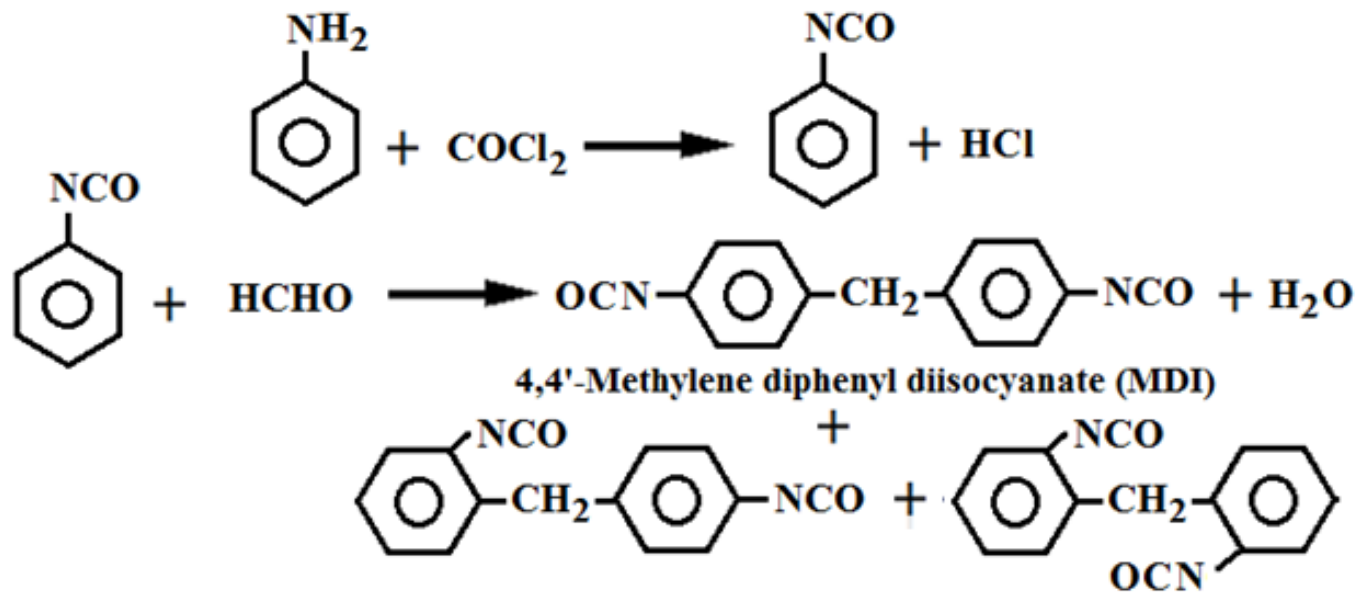
Nitrobenzene Market size was valued at US\$ 10.34 Bn. in 2021 and the total Nitrobenzene revenue is expected to grow at 5.8% from 2022 to 2027, reaching nearly US\$ 14.5 Bn.

Aniline manufacturing makes up the bulk of nitrobenzene applications, accounting for more than 95% of the total.

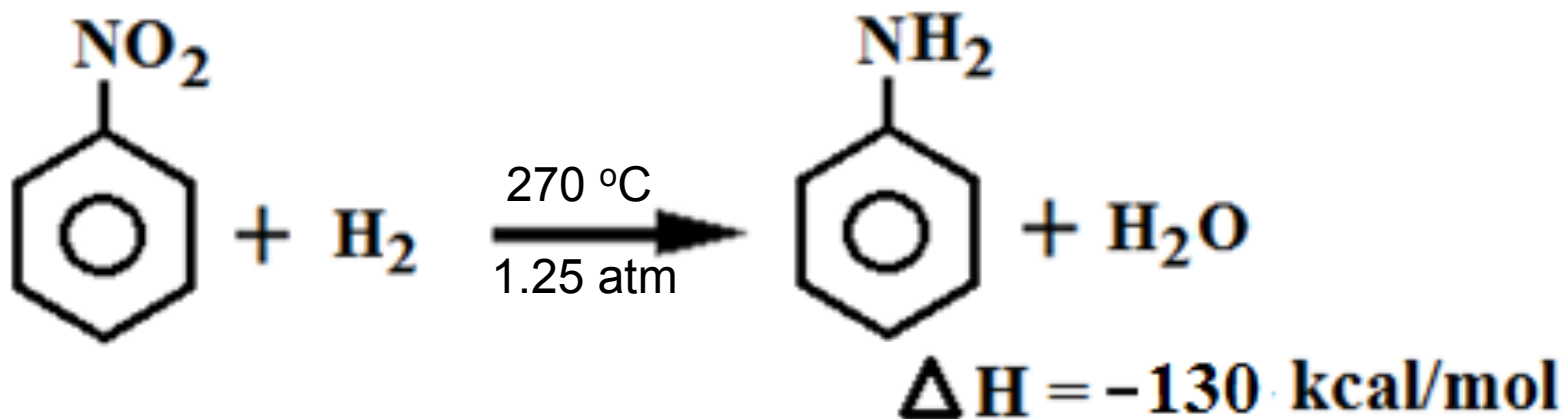
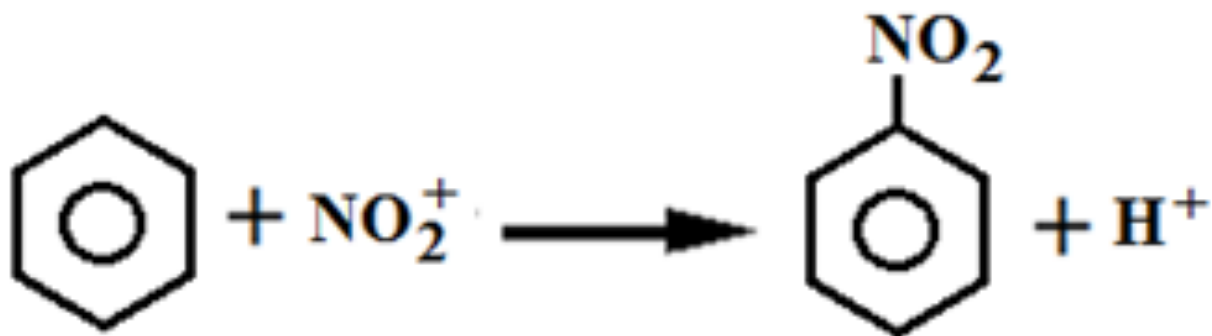
The aniline-derived methylene diphenyl di-isocyanate (MDI) serves as a primary predecessor for polymers utilized in a variety of end-user sectors, notably architecture, automobile, and many others. Additionally, polyurethane foams made from MDI, which are primarily applied in different structural insulation applications, are one of the most crucial characteristics in architecture, both in their ductile and brittle forms.

Aside from aniline, the substance is also used as an insecticide ingredient.

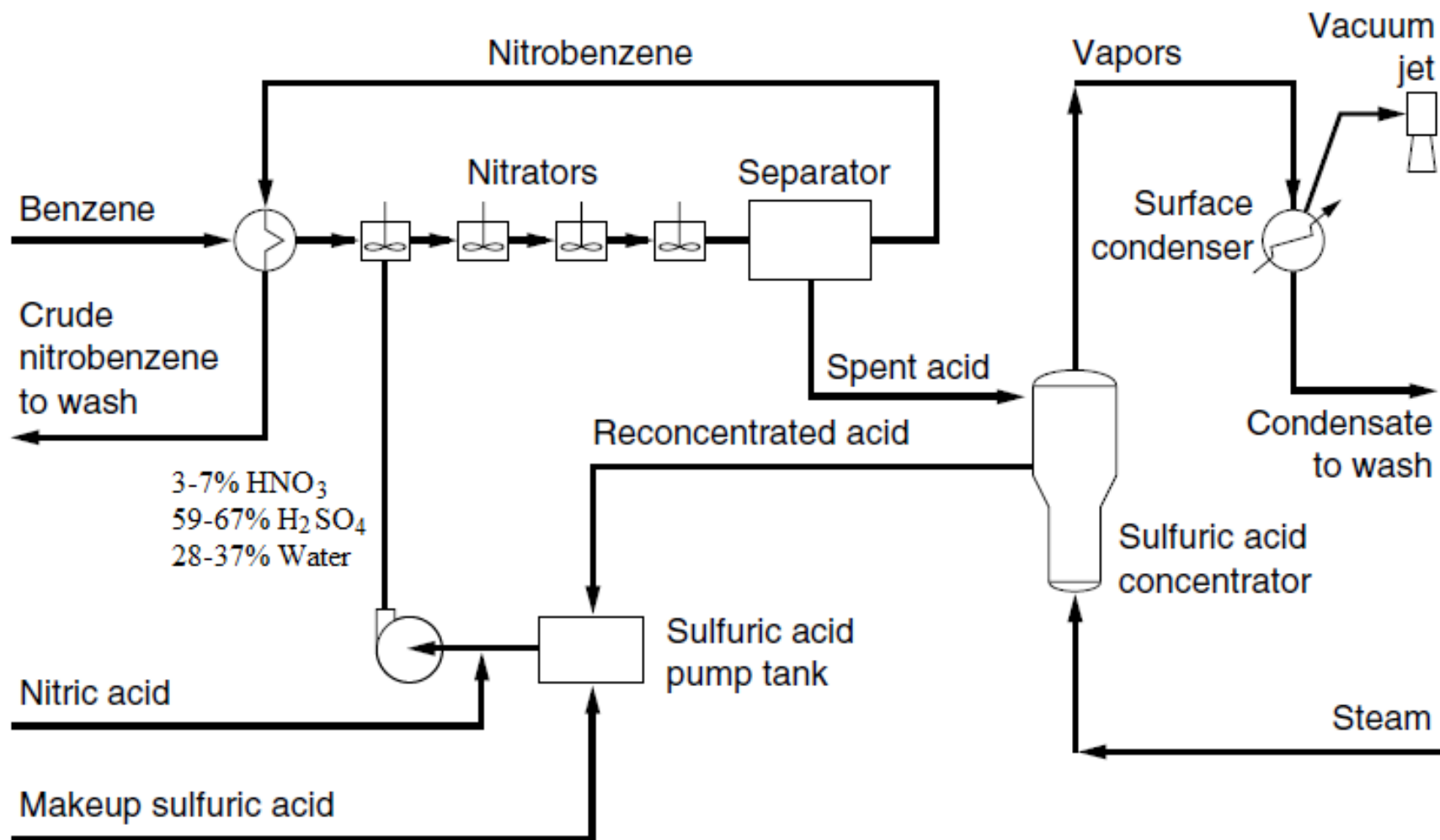
Nitrobenzene is used in the pharma industry to make paracetamol, the most widely used drug on the planet. Because paracetamol is mostly produced in India and China, APAC is the pharma industry's biggest user of aniline and hence nitrobenzene.



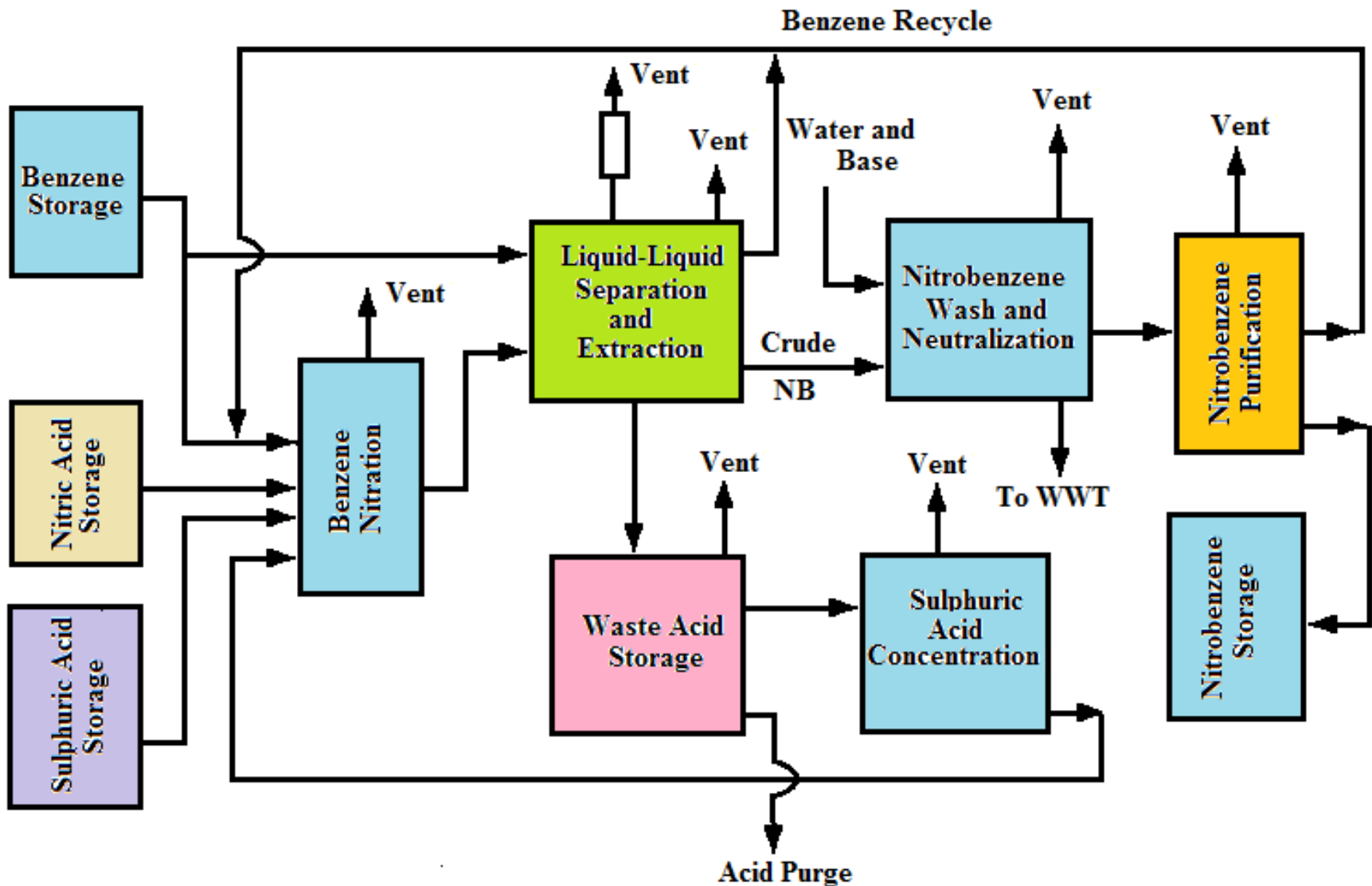
Polyurethane



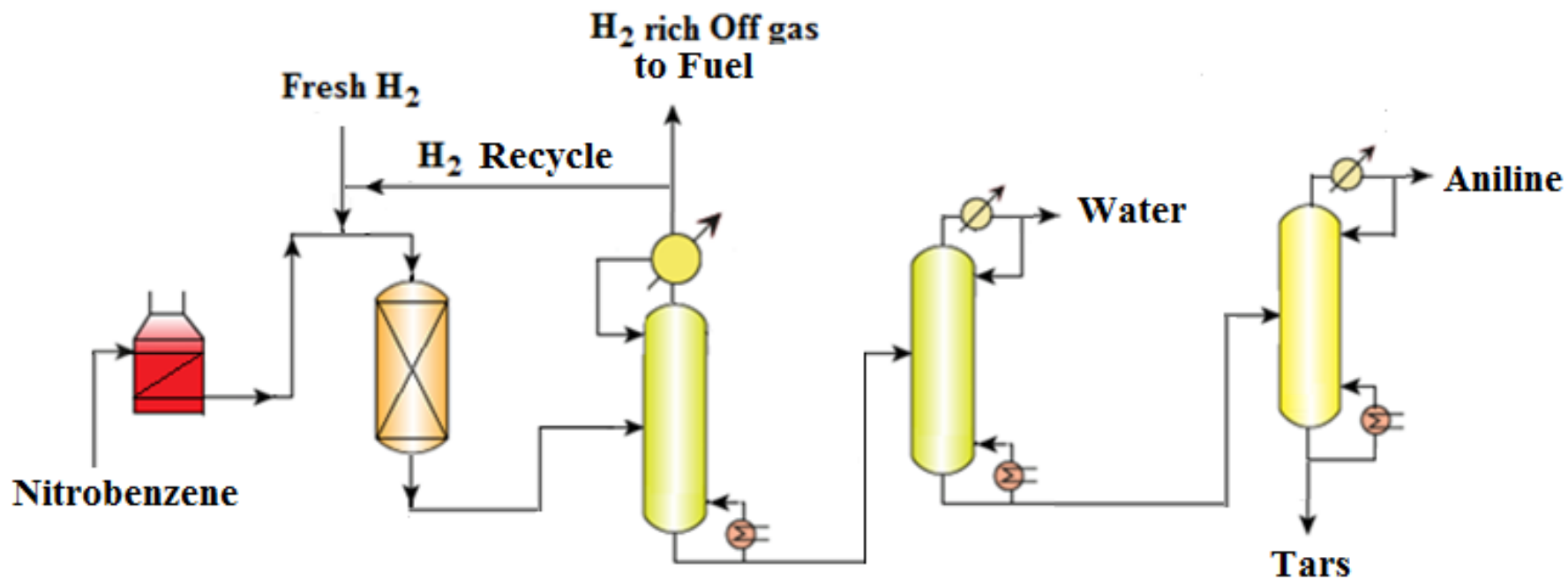
Catalyst: Cu or Pd on activated carbon or Alumina in combination with other metals (Pb, V, P and Cr) as modifiers/promoters



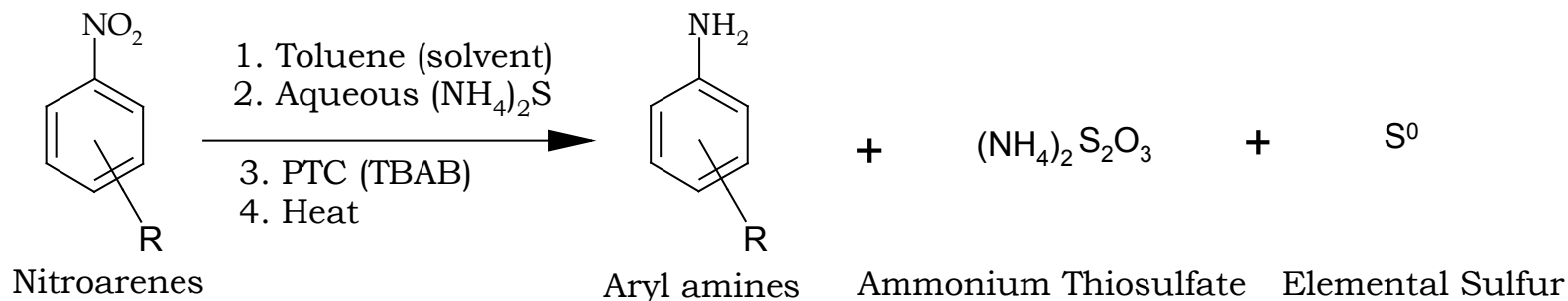
Flowsheet for the Production of Nitrobenzene



Block Diagram of Nitrobenzene Production



Production of Aniline from Nitrobenzene



where, R=CH₃ (Nitrotoluenes) or Cl (Chloronitrobenzenes)

Stoichiometry

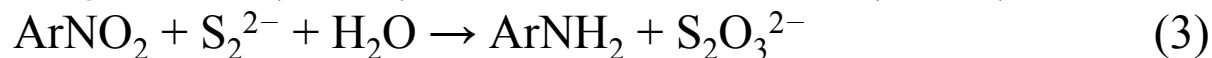
❖ Dauben (1973):



❖ Gilman (1941); Lucas and Scudder (1928); Meindl et al. (1984):



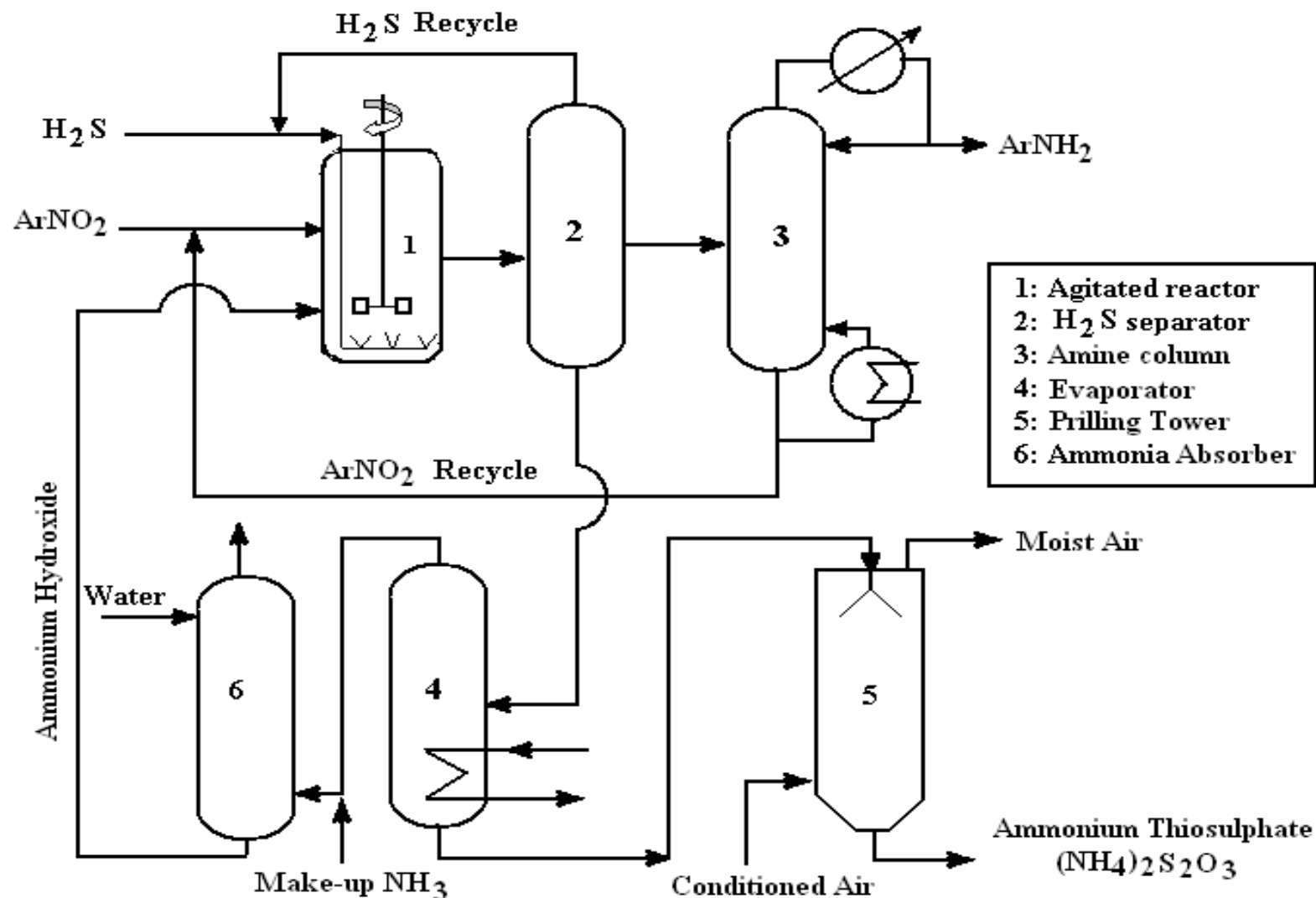
❖ Hojo et al. (1960); Bhave and Sharma (1981):



Maity et al., *Appl. Catal. A: General*, **2006**, 30 (2), 251-258.

Maity et al., *Appl. Catal. B: Environmental* **2008**, 77(3-4), 418-426.

Jeeru et al., *Indian Chem. Engr.* **2016**, 58(3), 279-296.

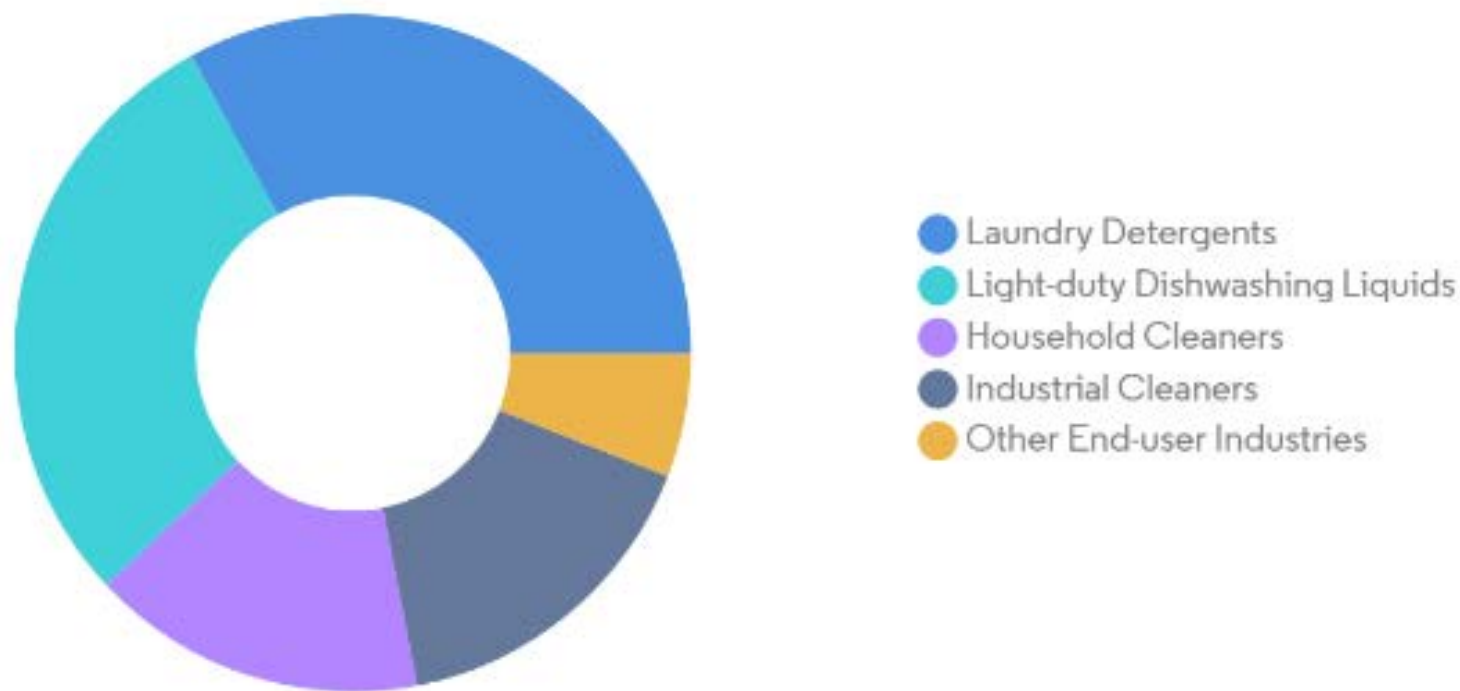


Process Flow Diagram for the Production of Aromatic Amines by Absorption of H_2S in Aqueous NH_3 Solution and Simultaneous Reaction with Aromatic Nitro-compounds

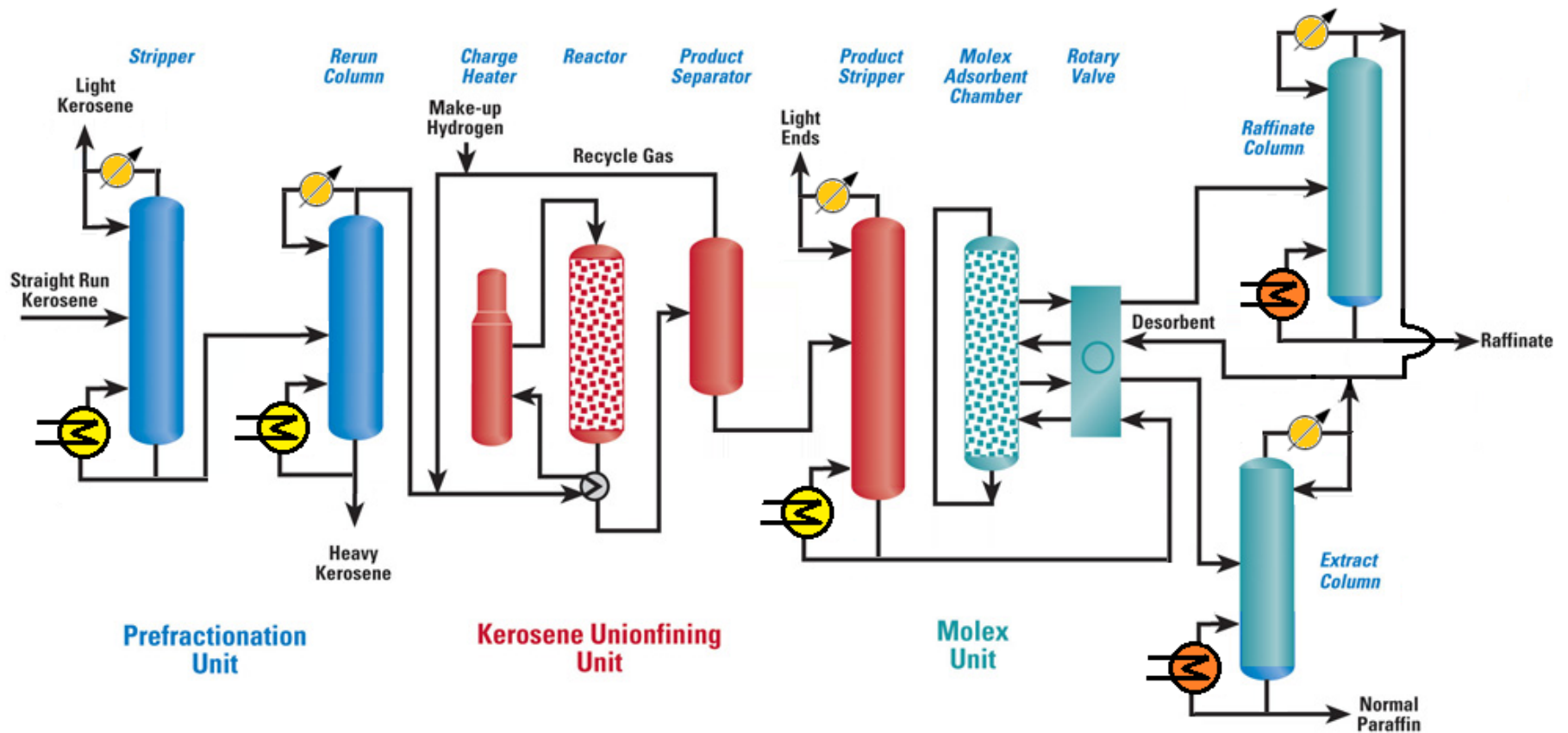
Production of
Linear Alkyl Benzene
(LAB)

The linear alkyl benzene market was valued at over 3,000 kiloton in 2021 and is projected to register a CAGR of less than 3% during the forecast period 2022-2027.

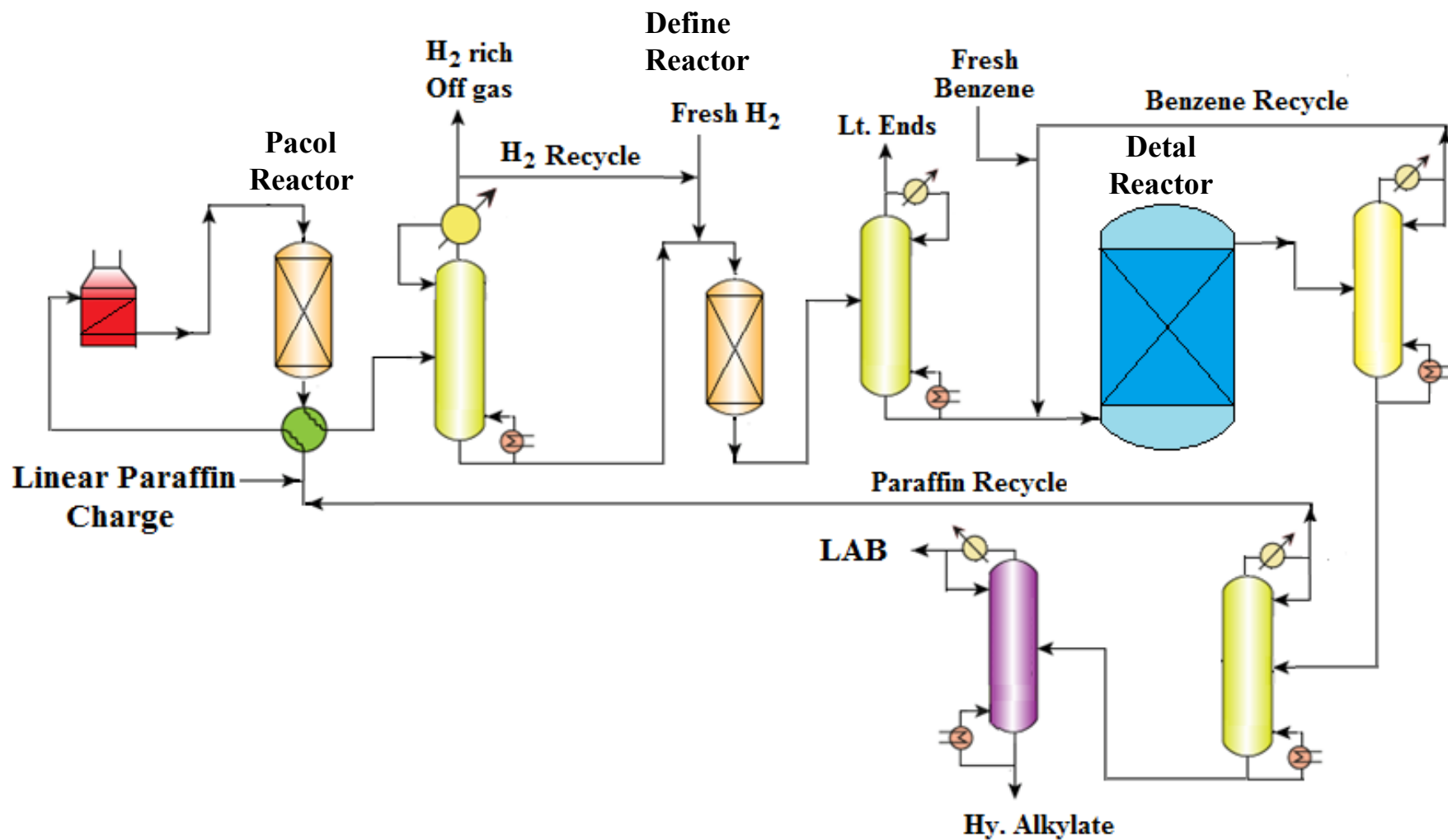
Linear Alkyl Benzene (LAB) Market, Volume Share (%), by End-user Industry, Global, 2021



A Review in Linear Alkylbenzene (LAB) Production Processes in the Petrochemical Industry. Aref Shokria and Safoora Karimia, Russian Journal of Applied Chemistry, 2021, Vol. 94, No. 11, pp. 1546–1559.



Flowsheet of Molex Complex



Production of Linear Alkyl Benzene (LAB)

Thank you!